Carnegie Mellon University
Dietrich College
Information Systems

Geographic Information Systems

California Fires

In this exercise, we create a map and a story map that illustrates the extent and dangers of recent California fires.

THE MAP

From the home page of your CMU ArcGIS Online account, create a new map (Map option at the top). A base map should appear. We next add various layers to visualize the extent of major California fires.

Select the 'Add layer' option and, instead of 'My content', select 'ArcGIS Online'. In the Search box, enter 'California fires'. A list of potential layers appears. Add the layer 'California Historical Fire Perimeters' by 'California Department of ...' to your map. The layer appears on your map and the properties panel appears on the right of your work area. Go ahead and close the properties panel.

Open the 'California Historical Fire Perimeters' layer. Note there are three layers contained within. We will examine recent fires. Hide the 'California Fire Perimeters (1950+)' layer and show the 'Recent Large Fire Perimeters (GT 5000 acres)'. Save your map with a descriptive name and, optionally, some useful tags. Your map will be available in the Content section of your account if you need to return to it.

Open the data table (... / Show table). Can you identify the five largest fires, by name, since 2020? (Hint: Sort the table on the 'GIS Calculated Acres') field. Many of these fires caused tremendous damage and loss of live. Select one of the largest fires, zoom in on it, and do a quick bit of background research on it. Prepare a few facts about this fire to present to the class - one minute, or less. (We'll call on a few students to present their findings.)

When you are happy with your work, again save your map. When you return to ArcGIS Online, you will find it under Content.

THE STORYMAP

Next, we create a StoryMap to illustrate your findings.

Under the 9-dot array at the top (next to your name), select StoryMaps. Under the Create option, select 'Story'. A blank template shows up.

Enter a descriptive story title (for example, maybe something like 'The Quarry Fire of 2023', with, of course, the name of your chosen fire.) Add a short introduction or subtitle.

Your work is constantly saved and will be available in the Content section of your account, and will be present as an option when you return to the StoryMap builder.

Download or link to a dramatic and relevant 'cover' image for your story map. Add it to your story map where it will appear at the top.

Under 'Design' select a Cover design and a Theme. Use one of the default themes or pick one from the Featured Themes gallery. Your theme should complement the story you intend to tell.

Wherever you see a plus sign in green, can add content of various types. At the first green plus sign (Tell your story), select Text, and add a brief paragraph or two explaining the situation. If you highlight your text, various formatting options appear.

Next, under the plus sign, select Map and add your California Fires map. If you have created other maps, they will also show up here. When you have selected your map, you can decide which layers you want to appear. Add some basic text below the map to explain it.

Select the Preview option at the top of the map to preview your work at any time. Choose the small x at the bottom right to return to the design/edit mode.

Optionally, add any other text, images, or other media that help your audience understand the story you are telling.

Note: If you make any changes to your original Cal Fires map in ArcGIS, these changes will be reflected in your StoryMap the next time you load it.